Laboratory Class 2: Body bouncing off a surface in a uniform gravitational field

Authors: P. Kowalczyk, A. Lamecki

The aim of this laboratory lesson is to familiarize the students with basic algebra (operations on vectors) and mathematical analysis (multivariable functions, partial derivatives).

1 Introduction

The problem concerns a body moving in a uniform gravitational field bouncing off an arbitraily defined, anaytic surface. When the body moves in a uniform gravitational field its acceleration is constant $\vec{a}(t) = [0, 0, -g]$ where $g = 10$ m/s^2. Then its velocity vector (antiderivative of acceleration) is

$$\begin{aligned} v_x(t) &= v_{0x}, \\ v_y(t) &= v_{0y}, \\ v_z(t) &= v_{0z} - gt, \end{aligned} \tag{1}$$

and its position can be expressed in the following form (antiderivative of velocity)

$$\begin{aligned} x(t) &= x_0 + v_{0x}t, \\ y(t) &= y_0 + v_{0y}t, \\ z(t) &= z_0 + v_{0z}t - \frac{1}{2}gt^2, \end{aligned} \tag{2}$$

where t represents time.

The impact point (collision with the surface) can be determined by combining the above equations with the given surface equation $z = f(x, y)$.

$$z(t) = f(x(t), y(t)) \tag{3}$$

where time of the impact $t = t_I$ first must be found. Then the impact point $P_I(x_I, y_I, z_I)$ is determined by

$$\begin{aligned} x_I &= x_0 + v_{0x}t_I, \\ y_I &= y_0 + v_{0y}t_I, \\ z_I &= z_0 + v_{0z}t_I - \frac{1}{2}gt_I^2. \end{aligned} \tag{4}$$

Note, that the equation (3) has usually more than one solution. So, we must reject (if they exist) the complex solutions, as well as the negative ones and zeros. If there is no proper solutions, there will be no collision.

For instance, if $z = x^2 + y^2$, then the relation (3) has the following form

$$z_0 + v_{0z}t - \frac{1}{2}gt^2 = (x_0 + v_{0x}t)^2 + (y_0 + v_{0y}t)^2 \tag{5}$$

and has no more than two real solutions. In the first step (the first impact), one of these solutions should be positive and the other should be negative. However, in the next

impacts one will be zero and the other will be positive. Therefore, it is sufficient to check if the solutions are real and choose the grater value.

Once the impact point is determined it becomes an initial point of the next movement. We can once again use the relations (3), but with the new values of initial conditions: $x'_0 = x_I$, $y'_0 = y_I$ and $z'_0 = z_I$. Next step is to determine the initial velocity of the body after bounce. As it was derived in the lectures (see: Lectures 2: Mathematics - basic modeling tool, slides 6-9), the relation between the incident velocity vector $\vec{v}_{(before)}$ and reflected velocity vector $\vec{v}_{(after)}$ has the following form

$$\vec{v}_{(after)} = \vec{v}_{(before)} - 2(\vec{v}_{(before)} \circ \vec{n})\vec{n} \tag{6}$$

where $\vec{n}$ is the unit vector normal to the surface at the point of collision. To calculate this vector we can use a simple formula (see: Lectures 3: Mathematics - basic modeling tool, slides 9-12):

$$\vec{n} = \frac{\vec{N}}{||\vec{N}||}, \qquad \vec{N} = \left[-\frac{\partial f}{\partial x}, -\frac{\partial f}{\partial y}, 1\right]. \tag{7}$$

The incident velocity vector can be determined from (2) for time $t = t_I$, as follows $\vec{v}_{(before)} = [v_{0x}, v_{0y}, v_{0z} - gt_I]$. Finally, the obtained reflected velocity becomes the initial velocity for the next movement $[v'_{0x}, v'_{0y}, v'_{0z}] = \vec{v}_{(after)}$.

If the collisions are perfectly elastic the process will never stop. However, in real life during the impact a part of the kinetic energy is absorbed. Let us assume that k represents the losses, so $k = E_{kin}^{(after)}/E_{kin}^{(before)}$. This effect can be taken into account in the calculation of the reflected velocity vector

$$\vec{v}_{(after)} = \sqrt{k}\left[\vec{v}_{(before)} - 2(\vec{v}_{(before)} \circ \vec{n})\vec{n}\right]. \tag{8}$$

2 Exercise

1. For a given data (analytic definition of the surface and initial conditions) write a Matlab program that simulates the first five bounces off of the body. Neglect the energy losses. Make a list of coordinates of impact points and the times between the sequential collisions. Make chart of the surface and put on it 5 trajectories. Show the obtained figure and the results to the teacher for approval and save it (see example in Fig. 1).

2. Calculate the kinetic $E_{kin} = mv^2/2$, potential $E_{pot} = mgz$ and total energy E_{total} for the impact points. Assume that the mass of the object $m = 1$ kg. Is the total energy constant? Collect the results in a table and show it to the teacher for approval.

3. Take into account the energy losses and repeat the point 2.1.

3 Report

Please, send:

- the code,
- the obtained figures (well described),
- the tables of results

to the given e-mail address. In e-mail title please indicate the student group affiliation!

Useful Matlab functions: roots, meshgrid, mesh, plot3, norm.

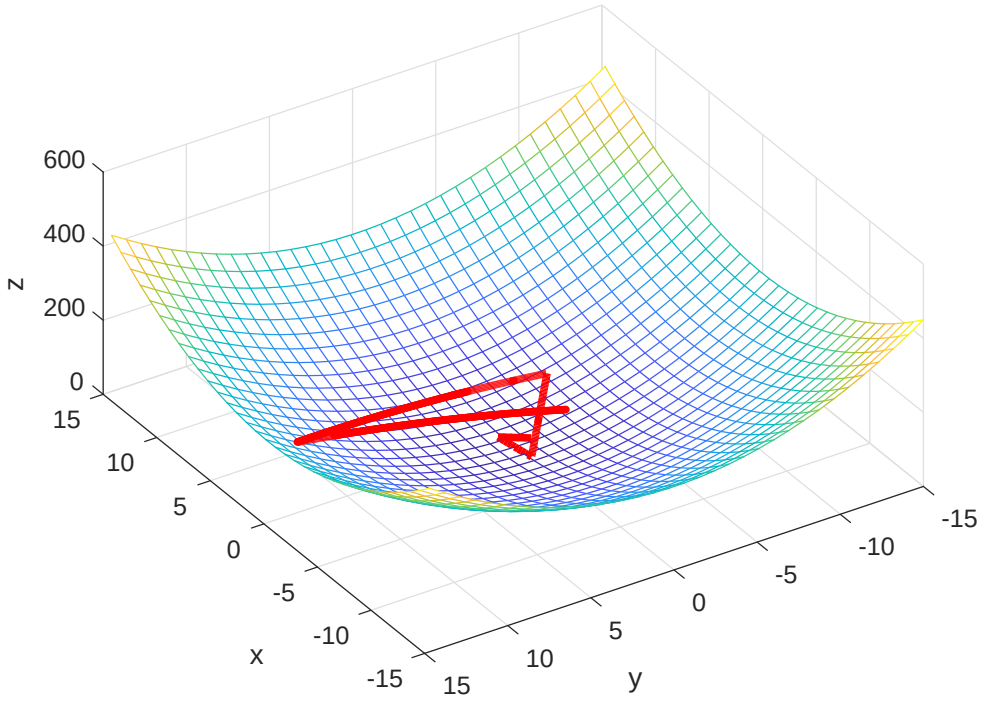


Figure 1: An example of the body movement

Example:
Data:
$f(x, y) = x^2 + y^2$,
$x_0 = -5$, $y_0 = 0$,$z_0 = 200$,$v_{0x} = 2$,$v_{0y} = 4$,$v_{0z} = 0$,
$k = 0.8$

Results (without losses - $k = 1$):

x_I	y_I	z_I	Δt	E_{kin}	E_{pot}	E_{total}
1.1516	12.3033	152.6967	3.0758	483.0327	1526.9673	2010.0000
3.7196	-5.1916	40.7882	2.5871	1602.1179	407.8821	2010.0000
-2.6942	0.6630	7.6981	0.5761	1933.0189	76.9811	2010.0000
1.1959	0.5497	1.7324	0.1137	1992.6760	17.3240	2010.0000
-0.5235	-0.8590	1.0119	0.0370	1999.8808	10.1192	2010.0000

Results (with losses - $k = 0.8$):

x_I	y_I	z_I	Δt	E_{kin}	E_{pot}	E_{total}
1.1516	12.3033	152.6967	3.0758	483.0327	1526.9673	2010.0000
3.6590	-4.7787	36.2244	2.8242	1551.1498	362.2437	1913.3935
-2.3182	0.3989	5.5330	0.6011	1547.8332	55.3303	1603.1635
0.9060	0.5192	1.0904	0.1094	1282.6929	10.9040	1293.5969
-0.4207	-0.8360	0.8759	0.0421	1028.2993	8.7590	1037.0583